



g w Wexford County LF
West Co

March 9, 2016

Project No. 16388377

Janice Heuer
MDEQ – Water Resources
Cadillac District Office
120 West Chapin Street
Cadillac, Michigan 49601-2158

**RE: WEXFORD COUNTY LANDFILL RULE 2210(Y)
EMERGENCY DISCHARGE REQUEST**

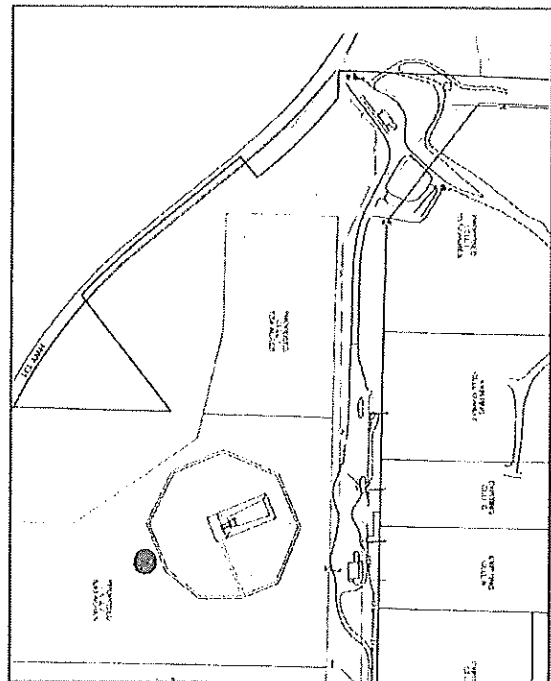
Dear Ms. Heuer:

On behalf of Wexford County Landfill, LLC (WCL), Golder Associates Inc. is requesting an emergency groundwater discharge authorization for wastewaters from the onsite leachate lagoon that is part of the onsite Leachate Evaporation System (LES). The Site is an active licensed 196.4 acre Type II solid waste disposal facility owned and operated by WCL. The Site is located at 990 Highway 131, Manton, Cedar Creek Township, Wexford County, Michigan. Specifically, the Site is located in the West ½ of the Northwest ¼ of Section 34 and the East ½ of Section 33, Township 23 North, Range 9 West, Cedar Creek Township, Wexford County, Michigan.

The LES is part of the Sites approved leachate collection, treatment and disposal system. It is a mobile device that converts leachate into a controllable aerosol, and eliminates the necessity to send leachate offsite to a wastewater treatment facility. The system separates the leachate into two phases, a water phase and a contaminant phase. The water phase is very light due to its small size (< 5 microns) and evaporates. The contaminant phase is much heavier and falls out within the constructed leachate lagoon. The LES is currently being used at another landfill in northern Michigan; that landfill has achieved a 100% reduction in leachate generation rates that need to be taken offsite for treatment and/or disposal.

The leachate lagoon was constructed in 2015 with approval for use granted on November 2, 2015 from the Office of Waste Management and Radiological Protection. WCL began routing leachate from the onsite leachate storage tank to the lagoon the following day, but quickly stopped due to the quantity of rainwater already in the lagoon and the projected precipitation for the remainder of the month. By flow estimates, approximately 90 percent of the wastewater currently in the lagoon is precipitation and not leachate from the Site.

Analytical data of the wastewater in the lagoon was collected at various times in December and most recently on March 2, 2016, a copy of which is attached. As shown on the summary table below, the current concentrations of constituents of concern are below Part 201 criteria for residential health drinking water criteria and below groundwater discharge criteria.



● Proposed Discharge Location

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Golder Associates Inc.
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Golder Associates: Operations in Africa, Asia, Australasia, Europe, North America and South America

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Hazardous Substance	Chemical Abstract Service Number	Residential Drinking Water Criteria	Nonresidential Drinking Water Criteria	Groundwater Criteria	Analytical Results				
					12/7/2015	12/10/2015	12/14/2015	12/21/2015	3/2/2016
Aluminum (B)	7429905	50 (V)	50 (V)	150					
Ammonia	7664417	10,000 (N)	10,000 (N)	5000	21000	26100	19100	22100	2700
Antimony	7440360	6.0 (A)	6.0 (A)		ND				
Arsenic	7440382	10 (A)	10 (A)		1				
Barium (B)	7440393	2,000 (A)	2,000 (A)		64				
Beryllium	7440417	4.0 (A)	4.0 (A)		ND				
Boron (B)	7440428	500 (F)	500 (F)		130				
Cadmium (B)	7440439	5.0 (A)	5.0 (A)		ND				
Chloride	16807006	2.6E+5 (E)	2.5E+5 (E)	250000	190000				
Chloroform	67663	80 (A,W)	80 (A,W)		14				
Chromium (B) (B,H)	16085831	100 (A)	100 (A)		3				
Cyba ¹	7440484	40	100		ND				
Copper (B)	7440508	1,000 (E)	1,000 (E)		4				
Iron (B)	7439896	300 (E)	300 (E)	300	100				
Lead (B)	7439921	4.0 (L)	4.0 (L)		ND				
Magnesium (B)	7439964	4.00E+05	1.10E+06		1.70E+04				
Manganese (B)	7439965	50 (E)	50 (E)	50	78	80	70	50	
Nickel (B)	7440020	100 (A)	100 (A)		ND				
Nitrate (B,N)	14797558	10,000 (A,N)	10,000 (A,N)		380	850	600	700	
Nitrite (B,N)	14797650	1,000 (A,N)	1,000 (A,N)	500	90	94	78	<150	
Phosphorus (Total)	7723140	83,000	2.40E+05	5.00E+03					
Selenium (B)	7782492	50 (A)	50 (A)		ND				
Silver (B)	7440224	34	98		ND				
Sodium	17341252	2.3E+5 (H+I)	3.60E+05	1.60E+05	1.10E+05				
Sulfate	14808798	2.6E+5 (E)	2.5E+5 (E)	250000	22000				
Zinc (B)	7440666	2,400	5,000 (E)		20				

Notes: 1. Aluminum and Phosphorus are not part of the leachate sampling requirements for the landfill under Part 115.
2. For Manganese, a concentration of 50 and 860 is established for aesthetic and health based criteria, respectively, under the Part 201 criteria tables for residential drinking water quality.

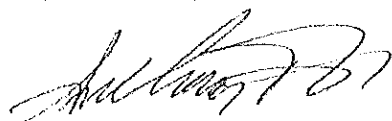
Pursuant to Rule 2210 – Items permitted to be discharged without a permit, of the Part 22 Rules of Act 451 a person may discharge wastewater if the proposed discharge meets the requirements of Rule 2204.

The proposed discharge location will be adjacent to the evaporator lagoon system which is greater than 100 feet from the property line as shown on the attached figure. No water supply wells are located near the discharge area. Analytical data collected of the wastewaters indicate that the discharge is less than Residential Drinking Water Quality concentrations outlined by Part 201 screening levels. A request to discharge pursuant to Rule 2210(y) is therefore being requested.

Rule 2210(y) authorizes "a discharge that has been determined by the department to have an insignificant potential to be injurious based on volume and constituents." The attached analytical data indicates that the discharge is below both residential criteria of Part 201 and below groundwater standards outlined in Rule 2222.

If you should have questions or would like to discuss in person, please contact Vicki Garon at 248-536-5437 (office) or 248-255-8280 (cell). Thank you in advance for your attention to this request.

GOLDER ASSOCIATES INC.



Anthony Testa
Staff Engineer



Vicki R. Garon, PE
Associate

cc: John Ozoga, MDEQ – OWMRP
Edward Ascione, WCL





RICK SNYDER
GOVERNOR

STATE OF MICHIGAN
DEPARTMENT OF ENVIRONMENTAL QUALITY
GAYLORD FIELD OFFICE



DAN WYANT
DIRECTOR

November 2, 2015

VIA E-MAIL

Mr. Eddie Ascione, Member
Wexford County Landfill, LLC
3947 North US 131
P.O. Box 1030
Kalkaska, Michigan 49646

Dear Mr. Ascione:

SUBJECT: Leachate Lagoon and Spray Curtain Engineering Certification and Operations Plan,
Wexford County Landfill, LLC, Waste Data System Number 470336

Staff from the Office of Waste Management and Radiological Protection of the Department of Environmental Quality (DEQ) has reviewed and hereby approves the Leachate Lagoon and Spray Curtain Certification dated October 2015 and the revised Leachate Recirculation Plan dated September 2015 for the Wexford County Landfill, LLC (Landfill). The DEQ's review was conducted under the provisions of Part 115, Solid Waste Management, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended; Michigan Compiled Law 324.11501 *et seq.*; and the rules promulgated thereunder.

Under this approval, the leachate disposal system (LDS) may be utilized within the approved containment area.

All related documents pertaining to the construction and operation of the LDS shall be placed in the Landfill's Operating Record in accordance with Part 115, Rule 438.

The DEQ reserves the right to require revisions to these approvals based on additional technical information or changes made in statutory regulations.

If you have any questions, please contact me at the number below or ozogaj@michigan.gov.

Sincerely,

John J. Ozoga, III
Senior Environmental Engineer
Office of Waste Management and Radiological
Protection
Gaylord Field Office
989-705-3403

cc/via e-mail: Mr. Don Suchocki, Landfill
Ms. Vicki Garon, Golder Associates, Inc.
Mr. Tris Waystack, Neptune Automated Wheel Wash System
Ms. Margie Ring/Mr. Philip Roycraft/Mr. Jim Staley, DEQ